Open Quantum: An Open Infrastructure Layer for Quantum Computing

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Abstract

Open Quantum is a platform that provides researchers and developers with access to quantum processing units (QPUs) from multiple hardware providers through a unified API and native integrations with widely used quantum programming frameworks, including Qiskit and PennyLane. The platform is operated by Quantum Rings—a team best known for their large-scale quantum circuit simulation—in partnership with qBitTensor Labs, which operates a decentralized quantum-compute network on the Bittensor protocol. Open Quantum offers two execution tiers: Public Compute, in which circuits are routed through the decentralized network, with free or subsidized access in exchange for the user's consent to the retention and study of execution data; and Private Compute, in which circuits execute on directly managed infrastructure and are deleted after execution. This paper describes the platform's architecture and execution model; the identity and verification mechanisms that connect a permissionless incentive network to metered quantum hardware; and the consented, cross-vendor corpus of real quantum workloads that the public tier accumulates, from which the platform will publish aggregate analyses beginning at the close of Summer 2026.

1 Introduction

Quantum computing promises solutions to problems in optimization, cryptography, and simulation that are intractable for classical computers. However, the high cost and limited availability of quantum processing units (QPUs), as well as the costly classical hardware requirements needed to run large-scale quantum simulators, have restricted access to a select few institutions and corporations. Accessing QPUs and simulators from a new provider also tends to require migrating algorithms and learning new systems and frameworks. These limitations have inhibited the growth of the quantum industry as a whole.

Quantum Rings is best known for their work on large-scale quantum simulation, which they have pursued since 2023 through the development of the Quantum Rings SDK: users can execute hundreds of logical qubits and millions of gate operations with high fidelity, all on consumer-grade classical hardware [1]. They first demonstrated their expertise in quantum simulation by simulating Google's 2019 quantum supremacy circuits on a classical computer with 32 GB of memory, achieving a record-breaking linear cross-entropy benchmarking (XEB) score of 0.622 [2]. They later revisited this experiment, running the same simulation on a single NVIDIA A100 GPU in 1:15:36, representing a 6.95×10^7 speedup compared to Google's

original classical prediction [3].

Open Quantum, launched in late 2025, extends this work from simulation to physical hardware. The platform provides access to QPUs from providers including IonQ, Rigetti, IQM, and AQT through a single API [4], with native plugins for the Qiskit [6] and PennyLane [7] frameworks and an integration with the qBraid development environment, so that existing programs target physical hardware with minimal modification.

The platform's Public Compute tier is offered in partnership with qBitTensor Labs [8], operator of the Quantum Compute subnet (SN48) on the blockchain-based Bittensor network. The subnet's incentive mechanism allows participating capacity providers to price circuit execution below the list price of equivalent direct access, which is what enables the platform's subsidized public tier. This paper is a revised edition of the launch white paper [5], updated to describe the platform as deployed. It covers the platform's execution model and architecture (Sections 2 and 3), the Open Quantum EDGE appliance for QPU operators (Section 4), the integration with the decentralized network (Section 5), the cross-vendor corpus of quantum workloads that the public tier accumulates (Section 6), and the platform's broader aims (Section 7).

2 The Open Quantum Platform

Open Quantum enables users to run quantum circuits on QPUs through a single API, eliminating the complexities of multi-vendor integrations: one account, one credential scheme, and one job model across all supported backends. Figure 1 summarizes this model—programs written in any supported framework reach any supported QPU through the same interface.

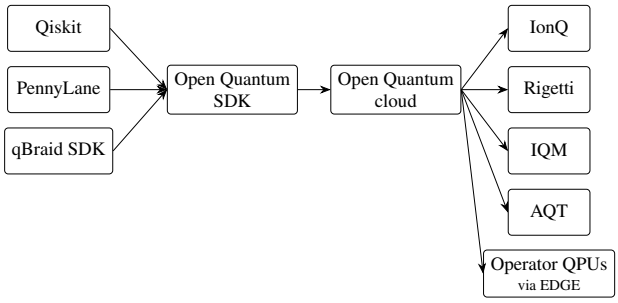


Figure 1: Many frameworks, many QPUs, one API. Programs written in Qiskit, PennyLane, or the qBraid environment reach any supported QPU—from providers including IonQ, Rigetti, IQM, and AQT, or operator systems connected through Open Quantum EDGE—with no vendor-specific integration.

Target audiences span traditional users, such as quantum researchers, academics, software developers, and enterprises, but also extend to students, enthusiasts, and innovators from adjacent fields, making quantum experimentation accessible without prohibitive costs.

2.1 Execution tiers

Every job is submitted under one of two execution plans.

Public Compute: Circuits are routed through the decentralized network operated by qBitTensor Labs. Access is free or subsidized: the incentive mechanism of Section 5 sustains pricing below the list price of equivalent direct access, and renewable no-cost credits allow a researcher or student to run circuits on physical QPUs without payment. In exchange, users consent to the retention and study of circuit and job data, which is used to improve calibration, scheduling, and platform performance and accumulates as the corpus described in Section 6; analyses from it are published only in aggregate, anonymized form. The platform's terms require attribution to the platform in publications resulting from work on the public tier; this paper is the canonical reference for that purpose.

Private Compute: Circuits execute on directly managed infrastructure operated through trusted, identified execution partners. Circuits and results are excluded from aggregate analysis, submitted circuits are deleted after execution, and jobs queue

on dedicated capacity. This tier is intended for proprietary workloads and production use.

Access is metered through prepaid credits of two kinds: *Spark credits*, granted at no cost on a renewing schedule and valid on Public Compute only, and *Full credits*, which are purchased and apply to either tier.

2.2 Interfaces

Users interact with the platform through a web portal or programmatically:

- **Core SDK**: a pip-installable Python package (`openquantum-sdk`) providing job preparation, submission, status polling, result retrieval, and per-job calibration-snapshot download. Circuits are accepted as OpenQASM 2.0 or 3.0, detected automatically.
- **Qiskit plugin**: implements the Qiskit `BackendV2` interface with `SamplerV2` and `EstimatorV2` primitives, and publishes per-backend transpilation targets (native gate sets, connectivity, timing) so that Qiskit's transpiler compiles directly to each device. Code written against the standard primitive interfaces runs without modification.
- **PennyLane plugin**: registers a native `openquantum.device` usable with the standard `qml.device` interface, including parameter-shift differentiation for variational workloads and support for counts, expectation-value, variance, probability, and sample measurements.
- **qBraid integration**: jobs may be submitted directly from the qBraid environment, which additionally provides cross-framework transpilation from other circuit representations.
- **MCP server**: a local Model Context Protocol [9] server (`openquantum-mcp`), built on the core SDK, exposes the platform to AI agents and agentic development environments. An agent can discover backends and credit balances, prepare a job and inspect its credit quote before any spend, submit circuits, poll for completion, and retrieve results and calibration data. Credentials remain on the user's machine, and paid submissions are guarded by explicit spend confirmation and a per-job cost cap.

Delegated authorization for third parties: External applications integrate through the same identity layer as the platform's own clients. Open Quantum exposes a standard OpenID Connect authorization-code flow (with PKCE), through which a user authorizes a third-party application—the qBraid environment, for example—to submit jobs on the user's behalf without ever sharing credentials with it. Access is consent-scoped and revocable: an application receives only the scopes granted at consent (job submission, or additionally account management for provisioning a new user's organization), access tokens are short-lived, refresh tokens rotate on use, and a user may revoke an individual application's authorization at any time without affecting their other sessions. This allows development environments, notebooks, and workflow tools to offer Open Quantum execution as a native capability while the platform retains a single point of identity, consent, and audit.

2.3 Hardware backends

The platform currently connects to trapped-ion systems from IonQ and AQT and superconducting systems from Rigetti and IQM; the available QPUs are listed in Table 1, with the short codes used to address them from the SDK and framework plugins. Each system links to its detail page, where full specifications are maintained. For every backend, the platform publishes the device's constraint metadata and its most recent calibration data, which users can download through the portal or the SDK; completed jobs may additionally carry a calibration snapshot taken at execution time. Backend availability evolves as providers and systems are added.

Table 1: QPU backends available on the platform at the time of writing, with their platform short codes. Each system links to its detail page at `www.openquantum.com/providers`.

System	Qubits	Backend short code
IonQ Forte-1	36	`ionq:forte-1`
IonQ Forte Enterprise	36	`ionq:forte-ent`
Rigetti Cepheus-1	108	`rigetti:cepheus-1-108q`
IQM Emerald	54	`iqm:emerald`
IQM Garnet	20	`iqm:garnet`
AQT IBEX Q1	12	`aqt:ibex-q1`

3 Platform Architecture

The platform separates user-facing scheduling from hardware-facing execution, with asynchronous queues between the two, so that clients, platform services, and execution sites scale independently.

- **Submission and preparation:** A client submits a circuit through the portal or an SDK. The circuit is stored in object storage and a preparation task is queued. A background worker validates the program, converts between OpenQASM versions (2.0 and 3.0) where a backend requires it, checks the circuit against the target backend's advertised constraints (qubit count, native operations, supported features), and computes a price quote for each available execution plan and queue priority. The user—or the SDK, automatically—selects a plan, at which point credits are reserved and the job enters the submission queue.
- **Execution:** Execution follows a pull model. Authenticated execution agents poll the platform's execution service for work destined for their backend; the platform never opens connections into a provider's network. A claimed execution is delivered as a compact descriptor containing a presigned reference to the circuit in object storage; the agent retrieves the circuit, executes it on the target device, and uploads results (and, where supported, a calibration snapshot taken at execution time) back to object storage. Failed executions are retried a bounded number of times before the job is marked failed and credits are refunded.
- **Execution sites:** Two kinds of agents execute jobs. Miners on the SN48 network (Section 5) operate agents backed by QPU capacity they own or procure, serving the public tier. Directly managed agents, operated through trusted, identified execution partners, serve the private tier. Operators can also connect systems they run themselves through Open Quantum EDGE (Section 4). In every case, new hardware is onboarded by adding backend metadata and deploying an agent, without change to the platform core.
- **Isolation:** Private Compute jobs are dispatched only to directly managed execution sites; circuit data is handled only by the platform and its vetted execution partners, and results return only through the platform's storage path.

4 Open Quantum EDGE

A quantum computer is rarely a single system. In practice it is a collection of loosely coupled subsystems—control electronics and pulse-level software, calibration and error-suppression tooling, cryogenic and environmental monitoring—each with its own interface, and none designed to be exposed to outside users. Open Quantum EDGE is an appliance that installs inside the operator's laboratory or data center and integrates with these components directly. To the outside world, it represents the instrument as a single, clean system on Open Quantum (Figure 2): jobs arrive as OpenQASM against the device's published constraints, execute through the local control stack, and return results and calibration snapshots through the platform's standard job model. Like every execution agent (Section 3), the appliance connects outward only—no inbound connections into the facility are required—and the operator retains full control of the instrument.

The appliance also carries the operational side of running a quantum system. It surfaces detailed telemetry from the dilution refrigerator and from the qubit controller through the platform, giving the operator live visibility into the instrument's state. Operators can schedule reserved windows, during which only designated organizations may access the system, and recurring maintenance windows, during which the backend is withdrawn from scheduling entirely.

An EDGE-connected system may be listed privately or publicly. Listed privately, the device is visible only to the operator's own organizations, giving in-house researchers the platform's standard interfaces—Qiskit, PennyLane, the Open Quantum SDK—over their own hardware, alongside the other backends they already use. Listed publicly, spare capacity is offered to the platform's users at a price the operator sets. Listing visibility is distinct from the execution tiers of Section 2: a listing governs who can discover and submit to the device, while the execution plan chosen per job governs its pricing and data handling, so jobs on a publicly listed system may still run under Private Compute plans. Sold capacity creates a stream of income for the operator, which accumulates as platform credits that the operator's researchers can spend on the other systems available through Open Quantum, or which can be remitted as revenue. Idle capacity thus converts into either reach—time on other modalities and devices—or income, while local users retain priority on their own machine.

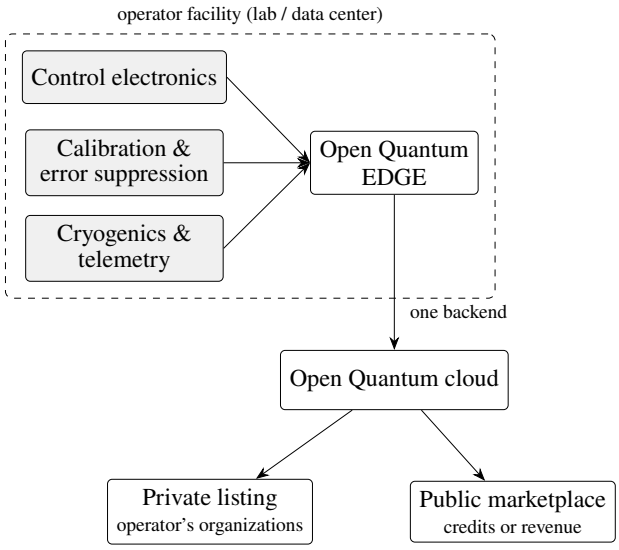


Figure 2: Open Quantum EDGE. The appliance installs in the operator's facility, integrates with the components of the QPU system, and represents the instrument as a single backend through Open Quantum. The operator lists the system privately, for their own organizations, or publicly on the marketplace, where execution accrues income as platform credits or remitted revenue.

5 Decentralized Compute in Partnership with qBitTensor Labs

The public tier's pricing derives from the platform's integration with the Bittensor ecosystem, a decentralized network in which independent subnets coordinate incentives with the TAO token [10]. Open Quantum's Public Compute is powered by the Quantum Compute subnet (SN48), operated by qBitTensor Labs (`https://www.qbittensorlabs.com`) [8], a company building quantum solutions within the Bittensor ecosystem. SN48 creates a two-sided market: demand arrives as circuits submitted through Open Quantum's API, and supply is provided by miners operating QPU capacity. Public jobs pass from the platform to qBitTensor Labs, whose validators assign each job to a suitable miner; the miner coordinates execution with the hardware provider, and results return along the same path (Figure 3). Because miners additionally earn TAO emissions in proportion to their assessed contribution, they can price execution below the list price of equivalent direct access; this subsidy is what sustains the public tier's below-list pricing and no-cost credit grants.

5.1 Identity and authorization

The subnet's participants are pseudonymous by design, while metered quantum hardware requires accountable access. The integration bridges the two with a token exchange: a miner or validator signs a request with its Bittensor hotkey; the platform verifies the signature, checks that the key is currently registered on the subnet by consulting the metagraph, distinguishes validators by their on-chain permit, and only then issues a scoped, short-lived OAuth2 token for the platform's execution APIs. Requests carry timestamps that expire within minutes.

5.2 Execution integrity

Two mechanisms tie the permissionless supply side to genuine quantum execution.

Onboarding verification: Capacity admitted to the network is vetted before it serves jobs: each

backend must be shown to be backed by a physical QPU from an identified hardware provider, with its device metadata (modality, qubit count, native operations, calibration characteristics) validated against the provider's published specifications.

Closed-loop job verification: Every job carries a verifiable execution trail. Because executions terminate at identified providers, the platform reconciles each reported execution against the provider's job records, confirming that the circuit was in fact submitted to, and executed on, the reported device. A job is complete only when this loop closes; discrepancies are surfaced before results are trusted or capacity is credited.

The subnet's incentive layer complements these checks: validators express their assessment of each miner's delivered capacity as on-chain weights, which determine the miner's share of TAO emissions, so sustained poor performance reduces a miner's reward without requiring any centrally administered penalty.

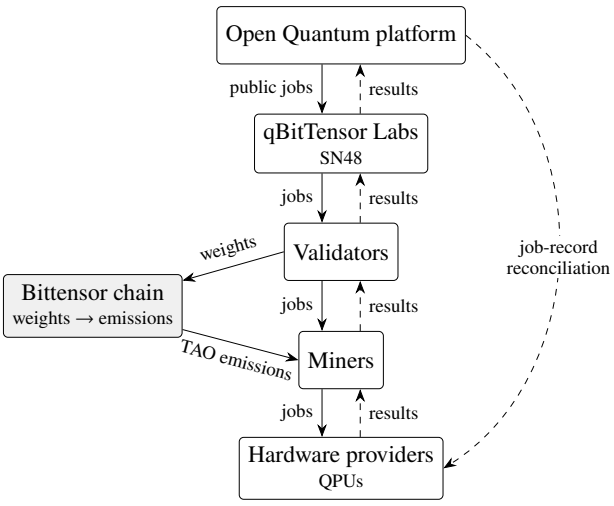


Figure 3: Execution and incentives on SN48. Public jobs pass from the platform to qBitTensor Labs and on through validators to miners, which execute them on provider QPUs; results return along the same path, and the platform reconciles each execution against the hardware provider's job records before the job closes. Independently, validators publish weights on-chain, which set each miner's share of TAO emissions.

5.3 Related subnet work

qBitTensor Labs also operates Enigma (SN63), a challenge platform that accumulates prize pools for adversarial problems in the same problem domain—including staged RSA-factoring milestones—with solutions published openly once claimed. Enigma is complementary to SN48: one funds standing quantum capacity, the other funds targeted attacks on the assumptions that capacity and its verification rest on.

6 A Public Corpus of Quantum Workloads

The public tier trades free and subsidized execution for consent to study the resulting data. That exchange accumulates a resource the field has lacked: a consented, cross-vendor record of real quantum workloads. For each public job the platform retains the submitted circuit itself, together with execution metadata (target backend, shot count, timing through the pipeline) and, where supported, a calibration snapshot of the device taken at execution time. Private-tier jobs are excluded entirely. Commercially sensitive work is the work most likely to run private, so the corpus over-represents exploratory, educational, and benchmarking use relative to the network as a whole, and analyses drawn from it must be read with that selection in mind.

Most accounts of what quantum computing is used for rest on surveys and self-declared intent. A workload corpus permits measurement instead: the composition of submitted work by algorithm family, recognized structurally from the circuits rather than from a form field; the distribution of circuit width and depth over time; how demand distributes across the devices, modalities, and price points of a heterogeneous fleet; and how queueing and execution time behave across systems.

The same corpus is a natural substrate for machine learning. Circuits labeled by structural family form training data for circuit classification; circuit–result pairs from devices with attached calibration snapshots support learned noise models and error-mitigation strategies; and submission

traces from the platform's MCP interface (Section 2) capture some of the earliest examples of AI agents preparing and submitting quantum workloads without a human in the loop. Because the corpus spans vendors and modalities, datasets derived from it carry a heterogeneity that no single-vendor source can offer.

The data gathered to date has reached critical mass, and beginning at the close of Summer 2026 the platform will publish seasonal aggregate analyses of network traffic as *Open Quantum Insights*: workload composition, circuit-shape trends, and system-level performance, reported in anonymized aggregate. Individual circuits, results, and identities are never published in these reports.

7 Goals and Aims

Open Quantum's aim is to broaden access to quantum computing:

- **Access**: Lower the cost and integration barriers to running programs on physical quantum hardware, for institutions and individuals alike.
- **Sustainability**: Through Bittensor's incentive mechanisms, reward contributed quantum capacity so that below-list public-tier pricing is sustained by the network's own economics rather than by indefinite subsidy or centralized funding.
- **Measurement**: Publish aggregate analyses from the public corpus (Section 6) so that the field's actual usage, and not just its forecasts, is visible to researchers, hardware providers, and the industry at large.

The long-term vision is quantum computing as accessible as cloud computing is today.

8 Conclusion

Open Quantum combines low-cost, uniform access to quantum hardware with a decentralized incentive layer that sustains that access economically. The platform's contribution is less any single component than the composition: standard framework interfaces in front, a provider-abstracted execution pipeline in the middle, and a verified, incentive-aligned supply network behind it. That composition also yields, as a byproduct of its consent-based public tier, a growing cross-vendor record of how quantum hardware is actually used. Publishing what that record shows, beginning with the first seasonal analyses at the close of Summer 2026, is part of the contribution the platform intends to make.

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